

Evidence of the Quantum Optical Nature of High-Harmonic Generation

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(Received 7 June 2024; revised 28 August 2024; accepted 30 September 2024; published 5 November 2024)

High-harmonic generation is a light up-conversion process occurring in a strong laser field, leading to coherent bursts of extreme ultrashort broadband radiation [Lewenstein *et al.*, Phys. Rev. A **49**, 2117 (1994)]. As a new perspective, we propose that ultrafast strong-field electronic or photonic processes such as high-harmonic generation can potentially generate nonclassical states of light well before the decoherence of the system occurs [Gorlach *et al.*, Nat. Commun. **11**, 4598 (2020); Stammer *et al.*, Phys. Rev. Lett. **128**, 123603 (2022)]. This could address fundamental challenges in quantum technology such as scalability, decoherence, or the generation of massively entangled states [Lewenstein *et al.*, Luca Argenti Michael Chini, 27 (2024)]. Here, we report experimental evidence of the nonclassical nature of the harmonic emission in several semiconductors excited by a femtosecond infrared laser. By investigating single- and double-beam intensity cross-correlation [Loudon, Rep. Prog. Phys. **43**, 913 (1980)], we measure characteristic nonclassical features in the single-photon statistics. We observe two-mode squeezing in the generated harmonic radiation, which depends on the laser intensity that governs the transition from super-Poissonian to Poissonian photon statistics. The measured violation of the Cauchy-Schwarz inequality realizes a direct test of multipartite entanglement in high-harmonic generation [Wasak, Phys. Rev. A **90**, 033616 (2014)]. This result is supported by the theory of multimodal detection and the Hamiltonian from which the effective squeezing modes of the harmonics can be derived [Gonoskov *et al.*, Phys. Rev. B **109**, 125110 (2024); Christ *et al.* New J. Phys. **13**, 033027 (2011)]. With this work, we show experimentally that high-harmonic generation is a new quantum bosonic platform that intrinsically produces nonclassical states of light with unique features such as multipartite broadband entanglement or multimode squeezing. The source operates at room temperature, using standard semiconductors and a standard commercial fiber laser, opening up new routes for the quantum industry, such as optical quantum computing, communication, and imaging.

DOI: [10.1103/PRXQuantum.5.040319](https://doi.org/10.1103/PRXQuantum.5.040319)

I. INTRODUCTION

The study of the quantum properties of light has led to significant advances in basic science and has had a

broad impact on modern technology. Since the groundbreaking experiment of Hanbury Brown and Twiss (HBT) [1] and the theoretical formulation of optical coherence by Glauber [2], highly sensitive measurements have helped to unravel the nonclassical properties of light and have provided access to novel methods of sensing [3], communication [4], and imaging [5–7]. In particular, single-photon emitters, squeezed states of light, and entanglement between photons are heavily studied from a fundamental and practical point of view [8]. The go-to sources to infer optical nonclassical correlations are nonlinear interactions

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of light with the anharmonic potential in crystals, atoms, and artificial atoms [9]. The platforms are mainly limited to the regime of low photon numbers in the visible spectrum, in which quantum effects are pronounced, limiting the applied perspectives. Recent experimental approaches have tried to overcome these limitations, e.g., by using high-photon-number squeezed states [10], which is crucial for transferring the technology from the laboratory to real-world applications. High-harmonic generation (HHG) in semiconductors using a compact fiber laser could lead to these unique properties. In addition to the intrinsic potential of quantum correlation in solid state systems, which could be transferred to HHG [11], the on-chip integration of such quantum optical systems is a major factor enabling future photonic quantum technologies.

In this paper, we experimentally investigate the quantum optical properties of HHG in semiconductors as a bright and spectrally broadband (exceeding multipetahertz) source of nonclassical light governed by the coherent strong field dynamics of electrons in the material [12–14]. Our studies use modest resources such as industrial-grade semiconductors, a commercial infrared fiber laser, and basic quantum optical detection. Briefly, HHG is an extremely nonlinear process, leading to the emission of radiation due to strong-field-driven recollision and acceleration of carriers in atoms, molecules, or solids [15–17]. When a strong laser field interacts with a medium, the potential of the material is periodically and anharmonically distorted on a suboptical-cycle time scale. Such ultrafast interaction leads to the emission of attosecond bursts of radiation that can be controlled before strong dephasing and decoherence channels (such as phonons in solids) occur. Depending on the symmetry of the system, one observes odd and even high-order harmonics of the fundamental driving-field frequency [18,19].

Recently, the nonclassical signatures in the infrared laser beam used to generate high harmonics in gases have been reported as optical Schrödinger cat states [20,21]. However, a direct measurement of the quantum optical nature of the high-harmonic emission has never been observed experimentally. Intense theoretical investigations have started to track the nonclassical features of the HHG process [12,21–23]. It has been predicted that a nonclassical driving field may alter the fundamental properties of the harmonic emission, leading, e.g., to a new path for the attosecond control of HHG light through squeezing [24]. In particular, the presence of multipartite entanglement between an arbitrary number of high-harmonic modes and squeezing effects has been predicted. Here, we investigate a reduced system with direct single-photon detection. We focus on the experimental study of the second-order intensity correlation function of two adjacent harmonics with quantum optical protocols derived from signal and idler photon statistics studies in the parametric down-conversion process (see Methods section in Supplementary

Material [25]) [26,27]. The measurement of intensity correlation functions is a powerful method to gain insight into the quantum properties of the source [28]. In particular, the second-order correlation function provides direct access to signatures of nonclassicality, an important step before realizing quantum state tomography [29].

II. MEASUREMENT OF NON-CLASSICAL CORRELATIONS

Here, we study the photon statistics of HHG photons from the interaction of a short-wavelength infrared (SWIR) ultrashort laser pulse with different semiconductors at room temperature. In detail, we investigate gallium arsenide (GaAs) with a [100] crystal cut, zinc oxide (ZnO [0001]), and silicon (Si [100]) for the third (H3) and fifth (H5) harmonics. The harmonic measurement, classification, and selection is shown in the Supplemental Material [25], where it is depicted in Fig. S7. The various band gaps of these semiconductors allow us to explore different strong-field regimes. The employed HBT-like setup, sketched in Fig. 1(a) and with more details in Fig. S1 of the Supplemental Material [25], measures the second-order intensity correlation function given by

$$g_{ij}^{(2)}(\tau) = \frac{\langle a_i^\dagger(\tau) a_j^\dagger(0) a_i(0) a_j(\tau) \rangle}{\langle a_i^\dagger(0) a_i(0) \rangle \langle a_j^\dagger(\tau) a_j(\tau) \rangle}, \quad (1)$$

where $a^\dagger$ and a denote the creation and annihilation operators of photons in mode i or j [22]. If $i = j$, the equation reduces to the standard single-mode normalized second-order intensity correlation function (ICF). We will refer also to this while omitting the indices. The brackets represent the average over an ensemble. Also, we adopt the notation $g_{ij}^{(2)}(0) = g_{ij}^{(2)}$.

In general, for a single-mode bosonic state, the value of $g^{(2)} = 1$ can be identified with Poissonian photon-number statistics, as obtained from coherent states. Further, $g^{(2)} > 1$ indicates super-Poissonian statistics and is connected to the bunched arrival of photons at the detector. Nonclassical effects such as superbunching result in $g^{(2)} > 2$, while $g^{(2)} < g^{(2)}(\tau)$ is connected to photon antibunching, as often obtained from single-photon sources. These values correspond to a violation of the bounds given by the Bose-Einstein ($g^{(2)} = 2$) and Poissonian ($g^{(2)} = 1$) photon-number distribution, respectively, as expected for a thermal or coherent source of light. Thus, both cases can indicate nonclassical photon statistics. While $g^{(2)} > 2$ is not a direct proof of nonclassicality, it is still a strong indicator, as superbunching can be observed only rarely from specially designed light sources. More commonly, this feature is obtained from nonclassical sources exhibiting collective emission [30] and squeezed states of light [31,32].

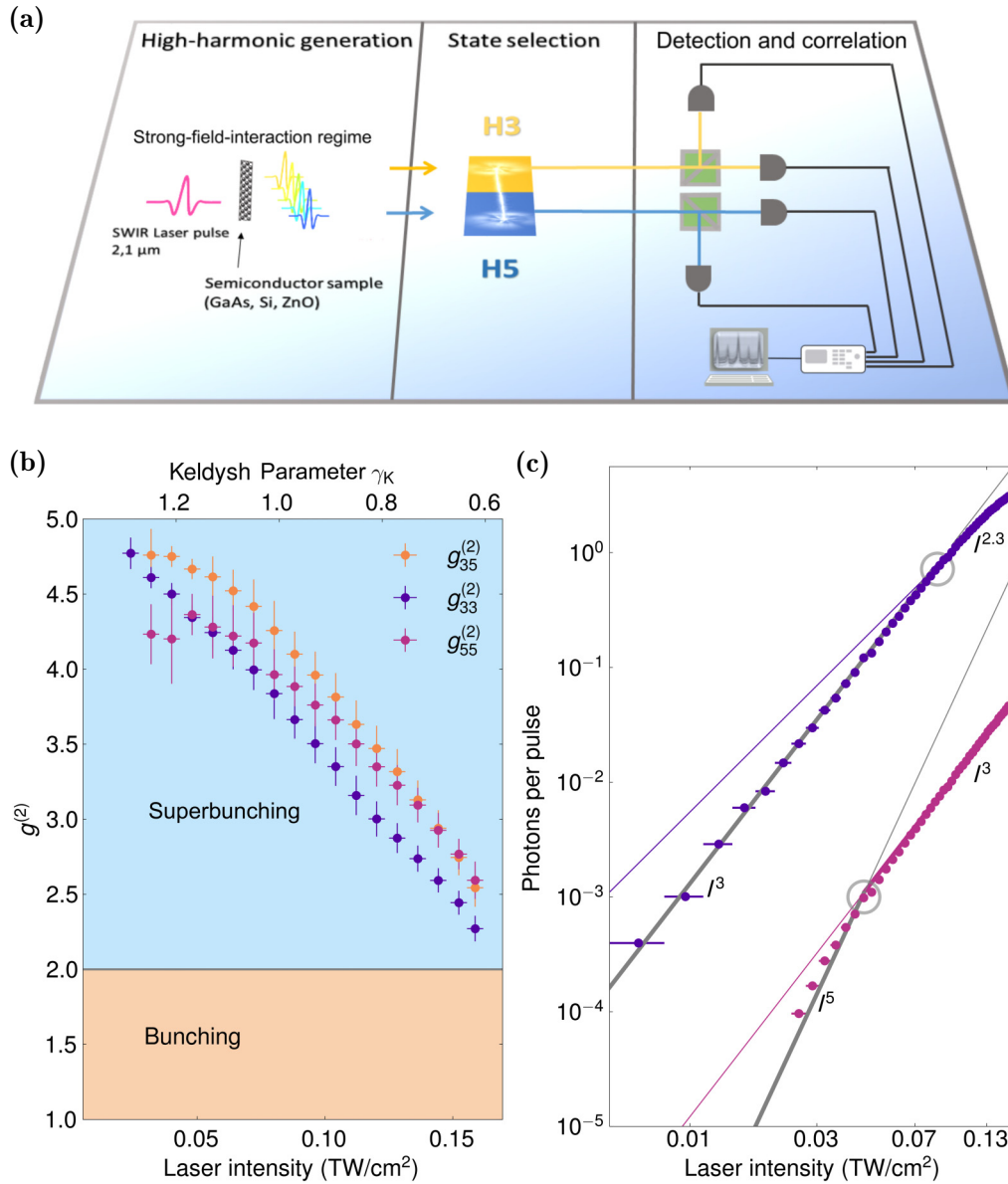


FIG. 1. The experimental scheme and the transition of the photon statistics and the nonlinear intensity dependence of high harmonics for the GaAs sample. (a) The conceptual diagram shows the different experimental stages. The high-harmonic generation process inside the semiconductor sample is driven by ultrafast SWIR laser pulses. Subsequently, the generated bosonic system is reduced to a bipartite system consisting of two harmonic orders, using optical filters for state selection (see the Supplemental Material [25]). The single-photon-resolving detection allows us to resolve the intensity fluctuations via an intensity-correlation measurement [33]. (b) The scaling of $g^{(2)}$ with the driving-laser intensity, obtained from the correlation of the high-harmonic photon counts generated in GaAs, measured in a single- and double-beam HBT setup. The blue-colored region indicates values of superbunching with $g^{(2)} > 2$. A decrease of $g^{(2)}$ with increasing laser intensity is recorded for correlations between the same order of high harmonics ($g_{33}^{(2)}$ and $g_{55}^{(2)}$) and for different high-harmonic orders ($g_{35}^{(2)}$). Interestingly, this behavior is connected to a change in the photon statistics. This behavior hints at a nonclassical state of light being generated. The error bars depict the standard error over repetition of the experiment. The scaling Keldysh parameter γ_K , given by $\gamma_K = \omega_L \sqrt{m^* E_g} / eE$, where ω_L is the angular frequency of the driving laser, m^* is the reduced electron mass, E_g is the band gap, E is the peak electric field, and e is the elementary charge [35], indicates the interaction regime. (c) The mean number of photons per pulse at the detectors in the HBT setup for H3 (blue) emitted at a central wavelength of 700 nm and at 420 nm for H5 (purple), calculated by correcting for the single-photon avalanche diode (SPAD) quantum efficiency and losses from optics. With an increase in the driving-laser intensity, a nonlinear increase of the photon number is observed. The deviation from the perturbative power scaling laws is marked with a circle at the intersection between the perturbative scaling and the best fit, where the exact-scaling power law is indicated on the figure.

As the time resolution of our system is well above the coherence time of the radiation, a time-integrated correlation function is measured in the described experiment. An estimated upper limit for the coherence time is the duration of the driving-laser pulse of 80 fs. Meanwhile, the timing resolution of the detection is on the order of tens of picoseconds and therefore

$$g^{(2)} = \frac{\int dt_1 \int dt_2 \langle \hat{a}^\dagger(t_1) \hat{a}^\dagger(t_2) \hat{a}(t_1) \hat{a}(t_2) \rangle}{\int dt_1 \langle \hat{a}^\dagger(t_1) \hat{a}(t_1) \rangle \int dt_2 \langle \hat{a}^\dagger(t_2) \hat{a}(t_2) \rangle} \quad (2)$$

is measured for the single-beam correlation [31,34]. Here, the indices refer to the two detectors. Transforming the $g^{(2)}$ function into the frequency domain by means of Fourier transformation,

$$g^{(2)} = \frac{\int d\omega_1 \int d\omega_2 \langle \hat{a}^\dagger(\omega_1) \hat{a}^\dagger(\omega_2) \hat{a}(\omega_1) \hat{a}(\omega_2) \rangle}{\int d\omega_1 \langle \hat{a}^\dagger(\omega_1) \hat{a}(\omega_1) \rangle \int d\omega_2 \langle \hat{a}^\dagger(\omega_2) \hat{a}(\omega_2) \rangle}. \quad (3)$$

Intuitively, this representation shows that a convolution of the photon statistics from all spectral modes impinging on the detector is measured. Analogously to the time domain, the finite timing resolution of the detectors does not allow them to discriminate the arrival time of the photon within the pulse.

We observe a clear dependence of $g^{(2)}$ on the pump power, with a scaling behavior displayed in Fig. 1(c) in the case of the GaAs semiconductor crystal. Each data point on the graph corresponds to the mean of multiple measurements and the error bars depict the corresponding standard error. For low driving-laser intensities, a distinct difference to the Poissonian photon-number distribution is found in the ICF (Fig. 1(b); see also Figs. S3(a) and S4(a) in the Supplemental Material [25]). The maximum $g^{(2)}$ values obtained from GaAs indicate superbunching with $g_{33}^{(2)} = 4.77 \pm 0.10$ and $g_{55}^{(2)} = 4.36 \pm 0.14$. The described characteristics are apparent for all investigated semiconductor crystals, while the slope as well as the exact maximum and minimum values for $g^{(2)}$ differ (Fig. 2).

Our measurements provide new insights into HHG, well beyond the classical Poissonian photon-number distribution as usually treated over the past three decades. This is interesting and drives fundamental hints. Indeed, HHG is an inherently coherent process, with the decoherence channels mainly governed by the electron dynamics of the system, which strongly depend on the laser intensity. Instead, for GaAs, ZnO, and Si, we observe values $g^{(2)} > 1$ for single- and double-beam correlations, indicating that the harmonic photons are bunched with super-Poissonian statistics. Still, the generation process appears to transition toward a Poissonian photon-number distribution as $g^{(2)} = 1$ is approached for higher driving-laser intensities. We interpret this observation as being due to the presence of two-mode squeezing in the high-harmonic radiation, as described below.

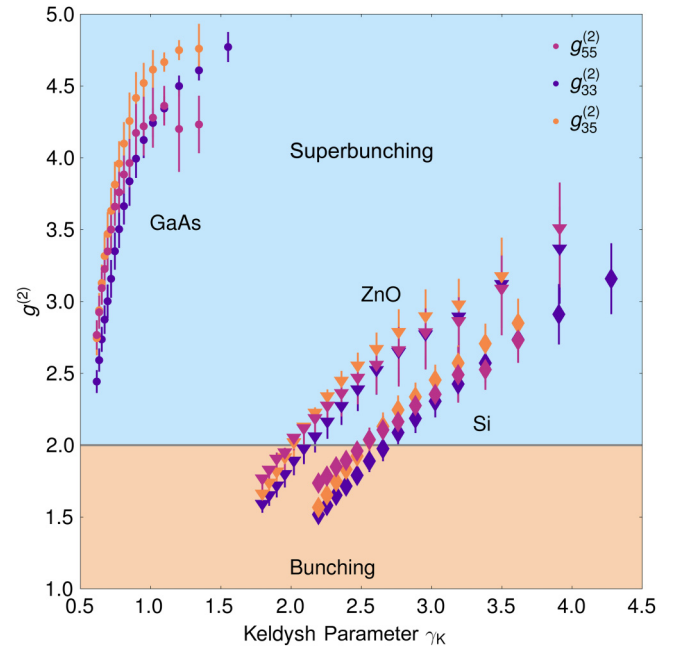


FIG. 2. Combined results from three different semiconductors. Experimentally obtained single- and double-beam second-order correlation values from the third and fifth high harmonics generated in three different crystals. The data are displayed as a function of the Keldysh parameter. The Keldysh parameter γ_K provides information about the strong-field interaction regime, taking into account the driving-laser intensity as well as the reduced electron mass and the band gap of the material. GaAs shows super-Poissonian statistics even at the lowest Keldysh parameter. Independent of the generation medium, a transition from super-Poissonian toward Poissonian photon-number statistics is observed, when the Keldysh parameter decreases (i.e., due to an increase in the driving-laser intensity). The transition in the photon statistics hints at the generation of a nonclassical state of light. The blue-colored region indicates values of superbunching with $g^{(2)} > 2$.

To characterize the harmonic source in terms of nonclassical properties, we evaluate the inequality

$$g_{ii}^{(2)} g_{jj}^{(2)} < [g_{ij}^{(2)}]^2 \quad (4)$$

for a multimode twin-beam state, where the indices i and j refer to the two different harmonic modes. If fulfilled, Eq. (4) indicates a nonclassical light source, as it corresponds to a violation of the Cauchy-Schwarz inequality (CSI), which is strictly valid for classical correlations [22,36]. It is well known that the violation of the CSI by the second-order correlation function indicates the presence of particle entanglement in a many-body system of bosons [37,38].

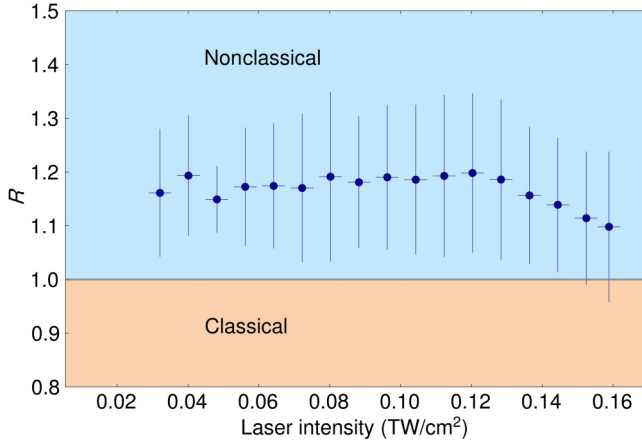


FIG. 3. The violation of the multimode Cauchy-Schwarz inequality. The graph shows the R parameter, defined as in Eq. (5), calculated from the experimental data for the GaAs sample, shown in Fig. 2. A value $R > 1$ corresponds to the violation of the classical limit given by the Cauchy-Schwarz inequality. The classical limit is depicted as the gray line. The blue-colored region indicates values of nonclassicality. The error bars are calculated by error propagation from the original data. The graph shows that the obtained correlation values cross the classical limit of 1, indicating nonclassical multimodal correlations. A violation is recorded within one standard deviation.

To this extent, we calculate the R parameter, defined as

$$R := \frac{[g_{ij}^{(2)}]^2}{g_{ii}^{(2)} g_{jj}^{(2)}}, \quad (5)$$

with the experimental data. According to the inequality in Eq. (4), the bipartite harmonic emission is nonclassical if $R > 1$. The results shown in Fig. 3 report the R parameter calculated from the experimental data in Fig. 1(a). The graph shows that the acquired correlation values violate the CSI within one standard deviation. A weaker violation is observed from the other samples (see Fig. S6 in the Supplemental Material [25]). Our interpretation is that the observed trend of the R parameter with increasing laser intensity is due to the increase of the number of photons in both harmonic modes [36].

The presence of squeezing and photon bunching has already been theoretically predicted for HHG in atomic targets [12,22,23] and strongly correlated materials [39], and in correlated many-body systems [40]. Expanding upon these prior studies, our theoretical analysis of HHG in semiconductors shows that harmonic modes can exhibit single-mode and two-mode squeezing. For our analysis, we adopt the derivation of the quantum properties of intraband harmonics studied in the interaction picture [14]. Here, we perform additional approximations specific to our experimental conditions. We include second-order terms with respect to the field operators for the harmonic modes

in the interaction Hamiltonian. This is motivated by an effective expansion of the Hamiltonian in terms of the ratio of the operator norm of the field operators of the harmonic fields over the operator norm of the fundamental laser mode. Furthermore, the system is reduced to a bipartite state $|\Psi\rangle_{35}$, consisting of the third and fifth harmonic modes, as these two harmonic modes dominate at the reported intensities.

III. THEORETICAL MODEL AND INTERPRETATION

In the interaction regime of the present study, the HHG process is mainly due to intraband excitations and interband transitions are neglected. This is well justified for HHG photons with energies below the band gap of the material, as is the case for the investigated ZnO and Si samples. For GaAs, photons stemming from interband contributions are expected to additionally contribute to the overall signal. It is not possible to clearly experimentally differentiate the magnitudes of the two processes. Since intraband excitations dominate the overall signal [41], the qualitative predictions made by the model using this approximation remain satisfactory. The single-particle Hamiltonian describing the interaction is written as

$$\hat{H} = n_e E_c \left[\hat{\vec{p}} - \frac{e}{c} \hat{\vec{A}} \right] + \omega_3 \hat{n}_3 + \omega_5 \hat{n}_5, \quad (6)$$

where e is the elementary charge, c is the speed of light, n_e is the time-independent number of electrons in the lowest conduction band, $\hat{\vec{p}}$ is the momentum of the electrons, $\hat{n}_j = \hat{a}_j^\dagger \hat{a}_j$ is the photon-number operator of harmonic mode j with frequency ω_j , E_c is the conduction-band dispersion, and $\hat{\vec{A}} = \hat{\vec{A}}_L + \hat{\vec{A}}_3 + \hat{\vec{A}}_5$ is the vector potential operator of the entire radiation field, where the undepleted driving-laser field is treated as a classical field. Treating the canonical momentum of the Bloch electrons as dominated by the driving-laser field and switching to the interaction picture yields the time-dependent interaction Hamiltonian

$$\hat{H}_I = n_e E_c \left[-\frac{e}{c} \hat{\vec{A}}(t) \right]. \quad (7)$$

We focus on the polarization projection, which allows us to treat the vector potential as a scalar quantity, and we approximate the band dispersion as a cosine potential having the analytic form $E_c(K) = E_g (1 - \cos(\pi K/K_c))$. Substituting the potential operator with the respective field operators, the second-order expansion around $-e/cA_L$ gives

$$\begin{aligned} \hat{H}_I(t) = n_e E_g \left\{ 1 - \cos\left(\frac{\pi \tilde{A}_L}{K_c}\right) + \sin\left(\frac{\pi \tilde{A}_L}{K_c}\right) \left[\tilde{c}_3 (\hat{a}_3(t) + \hat{a}_3^\dagger(t)) + \tilde{c}_5 (\hat{a}_5(t) + \hat{a}_5^\dagger(t)) \right] \right. \\ \left. + \frac{1}{2} \cos\left(\frac{\pi \tilde{A}_L}{K_c}\right) \left[\tilde{c}_3^2 (\hat{a}_3(t) + \hat{a}_3^\dagger(t))^2 + \tilde{c}_5^2 (\hat{a}_5(t) + \hat{a}_5^\dagger(t))^2 + 2\tilde{c}_3\tilde{c}_5 (\hat{a}_3(t) + \hat{a}_3^\dagger(t)) (\hat{a}_5(t) + \hat{a}_5^\dagger(t)) \right] \right\}, \quad (8) \end{aligned}$$

where $\hat{a}_j^\dagger(t)$ and $\hat{a}_j(t)$ are the creation and annihilation operators, respectively, of high-harmonic mode $j = 3, 5$. Further, $\tilde{c}_j = \pi/K_c \sqrt{\pi e^2/\omega_j V}$, in which K_c is an inverse lattice constant, E_g is the conduction-band half-width, and V is the quantization volume. The classical laser potential is denoted as $\tilde{A}_L = e/cA_L$. An expansion of the two last terms having quadratic dependencies of the field operators in Eq. (8), shows the equivalence to an interaction Hamiltonian inducing single- or two-mode squeezing in the harmonic modes. Based on our theoretical model, there are intervals in which the squeezing terms dominate the HHG interaction process, while at higher intensities, effects induced by terms linear in the field operators should become stronger. This hints at an optimal operation regime for the source, in which the quantum optical effects are pronounced and could then also occur at higher electric field strength. A detailed analysis of Eq. (8) can be found in the Supplemental Material [25].

Combining our three main results, the presence of power-dependent $g^{(2)}$ values in both single- and double-beam correlations (Fig. 2), as well as violation of the Cauchy-Schwarz inequality (Fig. 3) and the derived interaction Hamiltonian (Eq. 8), we associate high-harmonic generation with a quantum state that exhibits squeezing.

Photon-counting experiments can resolve signatures of squeezing effects, as squeezed states of light characteristically display a change in photon-number statistics depending on the squeezing strength and the number of squeezing modes, as reported in previous work [31,32,42]. The superbunching for the measured high harmonics is consistent with related theoretical predictions [12]. The observed transition in the photon statistics is a combined result of the dependence of $g_{ij}^{(2)}$ on the squeezing strength [32,43] and the multimode broadband detection over the full harmonic line width [31]. When the driving-laser intensity is increased, the squeezing strength and the number of occupied squeezing modes increases, resulting in $g^{(2)} \rightarrow 1$, as the detection does not discriminate between the contributions of the different modes [31]. Therefore, a convolution of photon statistics from different modes is effectively measured. As the individual modes are not necessarily correlated, this results in a random detection of events with Poissonian statistics. The similarity of the scaling behavior is induced by the dependence of the single- and double-beam correlation on the squeezer distribution given by the product of the spectral-mode distribution and the optical

gain. In the low-gain regime, the value of the single-beam correlations depends on the effective number of squeezing modes, whereas the double-beam correlation is sensitive to the optical gain of the process [31]. These dependencies are the reason for the similar scaling of the measured correlation function between the probed semiconductors. The apparent reason for this is that the generation in GaAs has a stronger nonperturbative action, illustrated by a lower Keldysh parameter compared to ZnO and Si (see Fig. S6 in the Supplemental Material [25]). Thus, squeezing effects are less pronounced for ZnO and Si, leading, overall, to fewer bunched photons and less correlation between the harmonic modes. This, besides the above-mentioned quantum optical effects, contributes to the scaling of $g^{(2)}$ (Fig. 2) and the trend of the R parameter toward the classical limit (Figs. 3 and S6). This finding could be used as a way to engineer the photon statistics of the high-harmonic emission using various semiconductor, doping, or interaction regimes.

IV. CONCLUSIONS

We have shown that high-harmonic generation can produce nonclassical states of light with unique features such as multipartite broadband entanglement or multimode squeezing. Multimodal nonclassical correlations have gained interest over the past decade due to the potential applications; however, only a few performing and scalable bosonic platforms exist. The HHG source operates at room temperature using standard semiconductors and with commercial femtosecond infrared lasers, opening up new routes for the quantum industry, such as quantum optical communication, imaging, and computing. Here, we have reported experimental evidence of the quantum optical nature of the high-harmonic emission in various band-gap semiconductors. On the basis of measuring the second-order intensity correlation, we have found evidence of a nonclassical state of light in the bipartite H3 and H5 harmonics system in GaAs, ZnO, and Si. Our experimental results identify two-mode squeezing, which is imprinted in the transition from super-Poissonian to Poissonian photon statistics and the violation of the Cauchy-Schwarz inequality. This finding has been supported by a theoretical treatment of HHG relative to intraband processes, in accordance with recent theoretical predictions. We have explained the observed change in the HHG photon statistics when increasing the driving-laser intensity

with a dependence of the photon statistics on the squeezing parameter and the influence of the multimode broadband detection. In the near future, HHG in the quantum optical regime with a repetition rate in the megahertz regime, as demonstrated here, will provide an ideal platform to produce multipartite entangled photonic systems with a high photon number and complex states of light. We have shown that the photon statistics depend on the laser intensity and the semiconductor nature and contain nonclassical features, which offers an engineering playground for future applications. Thus, the quantum optical effects cannot be neglected and they hint at the possibility of detecting nonclassical features at higher intensities and higher harmonic orders. With such properties, high-harmonic generation, as a “quantum newcomer,” has interest both from an academic and an industrial perspective, together with the potential to initiate applications in quantum information, communication, and sensing.

ACKNOWLEDGMENTS

We acknowledge support from Benoît Deveaud, at the department of research of École polytechnique, and from Philippe Zeitoun and Stéphane Sebban, at the Laboratoire d’Optique Appliquée. We acknowledge fruitful discussions with Hugo Defienne from the Sorbonne University. H.M. acknowledges financial support from the European Innovation Council contract EIC open “NanoXCAN” (Grant No. 101047223) and Architects of Next Generation Communication (ATTOCOM) and Quantum diffractive Nanoscale Microscopy contracts from the Agence Nationale de la Recherche (ANR). J.B. acknowledges financial support from the European Research Council (ERC) for ERC Advanced Grant “TRANSFORMER” (Grant No. 788218), ERC Proof of Concept Grant “miniX” (Grant No. 840010), FET OPEN “PETACOM” (Grant No. 829153), FET OPEN “OPTologic” (Grant No. 899794), and FET OPEN “TwistedNano” (Grant No. 101046424), from Laserlab-Europe (Grant No. 871124), from the Marie Skłodowska-Curie Innovative Training Networks (ITN) “smart-X” (Grant No. 860553), from the Ministerio de Economía, Comercio y Empresa (MINECO) for Plan Nacional PID2020–112664 GB-I00, from the Agencia de Gestió d’Ajuts Universitaris i de Recerca (AGAUR) for 2017 SGR 1639, from MINECO for “Severo Ochoa” (CEX2019-000910-S), from the Fundació Cellex Barcelona, from the Centres de Recerca de Catalunya (CERCA) Program–Generalitat de Catalunya, and from the Alexander von Humboldt Foundation for the Friedrich Wilhelm Bessel Prize. This work was funded by the Deutsche Forschungsgemeinschaft (German Research Foundation, DFG) under Grant No. KO 3798/11-1. M.K. acknowledges support from the DFG under Germany’s Excellence Strategy within the Cluster of Excellence

PhoenixD (EXC 2122, Project ID 390833453) and Quantum Frontiers (EXC-2123, Project ID 390837967).

H.M. and D.T. conceived the study. D.T., V.C., V.Sh., and S.F. performed the experiments. M.K. and K.A.W. provided support on single-photon detection. D.T. analyzed the data and provided the plots. R.S. and D.T. conducted the theoretical analysis. D.T. and H.M. wrote the manuscript, with input from all authors. All authors discussed the results.

The authors declare no competing financial interests.

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